Survey on "Security Rules for AI-based System Architectures" (PART 2)

Thank you for your interest!

This online survey is part of an empirical study conducted by the [redacted]. The goal of this study is to gain insights into the use of architectural rules (i.e., best practices) to enhance the security of systems based on Artificial Intelligence (AI) components. Particularly, into practitioners' views regarding the adoption and conformance checking of those rules. The study is divided into two parts (rewarded separately) and should take, altogether, around 13 min.

PART 1: Technical Assessment (**0.3 GBP reward**)

- It consists of 3 generic questions about programming, 1 related to software security and 1 to ML/AI.
- It takes around 3 min.
- Your answers will determine **if you meet our criteria** to undergo the 2nd part of the study.
- If you meet our participation criteria, you will be **invited to take the second part** of the survey.

PART 2: Main Survey (**1.30 GBP reward**)

- A set of questions about your experience securing AI-based systems.
- It takes around 8 min.

IMPORTANT:

1. To participate in this study, you must be at least 18 years old.
2. Please read the **Participant Information Sheet** before starting with the survey. The sheet can be accessed via this link: [redacted]

If you would like to proceed with the **Main Survey (PART 2)**, please answer the following questions by placing a check mark accordingly.

(Tip: Select all that apply)

Question attribute	Value
other_position	end
save_as_default	N

I am age 18 or older

I have read and understood the Participant Information Sheet

I agree to participate in this research and I want to continue with the survey

(1) DEMOGRAPHICS

► Please answer the following demographic questions.

Please indicate your **age**:

Question attribute	Value
hide_tip	1
min_num_value_n	**18**

num_value_int_only	1

Please indicate your **gender**:

Question attribute	Value
answer_order	normal
hide_tip	1
other_position	default
other	Y

Other

Male

Female

Non-binary

Please select your **country of residence**:

Question attribute	Value
answer_order	normal
hide_tip	1
other_position	default
other	Y

Other

Afghanistan

Aland Islands

Albania

Algeria

American Samoa

Andorra

Angola

Anguilla

Antarctica

Antigua and Barbuda

Argentina

Armenia

Aruba

Australia

Austria

Azerbaijan

Bahamas

Bahrain

Bangladesh

Barbados

Belarus

Belgium

Belize

Benin
Bermuda
Bhutan
Bolivia (Plurinational State of)
Bonaire, Sint Eustatius and Saba
Bosnia and Herzegovina
Botswana
Bouvet Island
Brazil
British Indian Ocean Territory
Brunei Darussalam
Bulgaria
Burkina Faso
Burundi
Cabo Verde
Cambodia
Cameroon
Canada
Cayman Islands
Central African Republic
Chad
Chile
China
Christmas Island
Cocos (Keeling) Islands
Colombia
Comoros
Congo
Congo (Democratic Republic of the)
Cook Islands
Costa Rica
Côte d'Ivoire
Croatia
Cuba
Curaçao
Cyprus
Czech Republic
Denmark
Djibouti
Dominica
Dominican Republic

Ecuador
Egypt
El Salvador
Equatorial Guinea
Eritrea
Estonia
Ethiopia
Falkland Islands (Malvinas)
Faroe Islands
Fiji
Finland
France
French Guiana
French Polynesia
French Southern Territories
Gabon
Gambia
Georgia
Germany
Ghana
Gibraltar
Greece
Greenland
Grenada
Guadeloupe
Guam
Guatemala
Guernsey
Guinea
Guinea-Bissau
Guyana
Haiti
Heard Island and McDonald Islands
Holy See
Honduras
Hong Kong
Hungary
Iceland
India
Indonesia
Iran (Islamic Republic of)

Iraq
Ireland
Isle of Man
Israel
Italy
Jamaica
Japan
Jersey
Jordan
Kazakhstan
Kenya
Kiribati
Korea (Democratic People's Republic of)
Korea (Republic of)
Kuwait
Kyrgyzstan
Lao People's Democratic Republic
Latvia
Lebanon
Lesotho
Liberia
Libya
Liechtenstein
Lithuania
Luxembourg
Macao
Macedonia (the former Yugoslav Republic of)
Madagascar
Malawi
Malaysia
Maldives
Mali
Malta
Marshall Islands
Martinique
Mauritania
Mauritius
Mayotte
Mexico
Micronesia (Federated States of)
Moldova (Republic of)

Monaco
Mongolia
Montenegro
Montserrat
Morocco
Mozambique
Myanmar
Namibia
Nauru
Nepal
Netherlands
New Caledonia
New Zealand
Nicaragua
Niger
Nigeria
Niue
Norfolk Island
Northern Mariana Islands
Norway
Oman
Pakistan
Palau
Palestinian Authority
Panama
Papua New Guinea
Paraguay
Peru
Philippines
Pitcairn
Poland
Portugal
Puerto Rico
Qatar
Réunion
Romania
Russian Federation
Rwanda
Saint Barthélemy
Saint Helena, Ascension and Tristan da Cunha
Saint Kitts and Nevis

Saint Lucia
Saint Martin (French part)
Saint Pierre and Miquelon
Saint Vincent and the Grenadines
Samoa
San Marino
Sao Tome and Principe
Saudi Arabia
Senegal
Serbia
Seychelles
Sierra Leone
Singapore
Sint Maarten (Dutch part)
Slovakia
Slovenia
Solomon Islands
Somalia
South Africa
South Georgia and the South Sandwich Islands
South Sudan
Spain
Sri Lanka
Sudan
Suriname
Svalbard and Jan Mayen
Swaziland
Sweden
Switzerland
Syrian Arab Republic
Taiwan, Province of China
Tajikistan
Tanzania, United Republic of
Thailand
Timor-Leste
Togo
Tokelau
Tonga
Trinidad and Tobago
Tunisia
Turkey

Turkmenistan
Turks and Caicos Islands
Tuvalu
Uganda
Ukraine
United Arab Emirates
United Kingdom of Great Britain and Northern Ireland
United States Minor Outlying Islands
United States of America
Uruguay
Uzbekistan
Vanuatu
Venezuela (Bolivarian Republic of)
Viet Nam
Virgin Islands (British)
Virgin Islands (U.S.)
Wallis and Futuna
Western Sahara
Yemen
Zambia
Zimbabwe
Please select your **highest educational level**: **Question attribute** / **Value** answer_order / normal hide_tip / 1 other_position / default
Primary school or less
High school or less
Some college
Undergraduate degree (BSc., BA.)
Graduate degree (MSc., Ph.D.)
How many years of experience do you have working in **Open Source Software**? **Question attribute** / **Value** answer_order / normal hide_tip / 1 other_position / default
< 1 year
1 to 2 years
2 to 3 years
3 to 4 years

4 to 5 years

> 5 years

How many years of experience do you have working in **Commercial (closed) Source Software**?

Question attribute	Value
answer_order	normal
hide_tip	1
other_position	default

< 1 year

1 to 2 years

2 to 3 years

3 to 4 years

4 to 5 years

> 5 years

How would you rate your **software security knowledge/skills**?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

VERY HIGH

ABOVE AVERAGE

AVERAGE

BELOW AVERAGE

VERY LOW

How would you rate your **expertise working with Artificial Intelligence (AI) systems**?

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

VERY HIGH

ABOVE AVERAGE

AVERAGE

BELOW AVERAGE

VERY LOW

SURVEY STRUCTURE

You will be asked to assess a total of **5 (five) rules** for securing AI-based systems. For each rule, you will be asked **6 (six) questions** about their importance, conformance, and applicability. The survey concludes with **1 (one) final question** about rules not covered in this study.

IMPORTANT: Please answer based on your experience!

Question attribute	Value
save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed."

Based on your experience, please indicate **how often** you have applied this **architectural rule** while working with AI systems.

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

NEVER

RARELY

SOMETIMES

OFTEN

ALWAYS

Based on your own experience, how could you check if an AI system **conforms to this rule** (i.e., does not violate it)?

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify
other_position	end
other	Y
save_as_default	N

Manually, following a structured or systematic approach (e.g., checklists).

Manually, not following any structured or systematic approach (intuitively).

(Semi-) automatically, tool-supported (e.g., static analysis).

Other

Which **sources/types of information** could you use to check this rule?

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify.
other_position	end
other	Y
save_as_default	N

Abstract representations of the system architecture (e.g., data flow diagrams).

Unstructured architectural information available across artefacts (e.g., code comments, code review comments).

Third-party documentation (e.g., API documentation).

Structured architectural information available as official documentation

Other

Can you explain how the **conformance check** for this rule would be **conducted**? (e.g., the overall strategy, use of sources, etc.)

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

Question attribute	Value
save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

Please indicate, based on your experience working with AI systems, how much do you **agree** or **disagree** with the following statements.

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

Question attribute	Value
save_as_default	N

I believe this rule helps create AI architectures with clearly defined and easy-to-maintain modules

I believe this rule helps reduce inconsistencies (e.g., unnecessary steps and errors) across the deployment pipeline of AI systems

I believe this rule contributes to the effective testing of AI applications and facilitates their transferability (e.g., handing them over to colleagues)

I believe this rule helps avoid unintended/unwanted accesses to the output of an AI module (e.g., from another AI module)

I believe this rule helps avoid dependency chains across AI models and the need for hotfixes to improve the accuracy of the final model

I believe this rule helps reduce the amount of additional/supporting code needed to get data into and out of AI modules (e.g., packages)

I believe this rule helps avoid having multiple AI components packed into a single big file

I believe this rule helps reduce the number of fixes/adjustments across multiple AI components when 3rd-party libraries get updated

I believe this rule helps reduce the number of unnecessary code fragments, methods or classes added to debug AI systems

COMPLETELY DISAGREE

DISAGREE

SOMEHOW DISAGREE

SOMEHOW AGREE

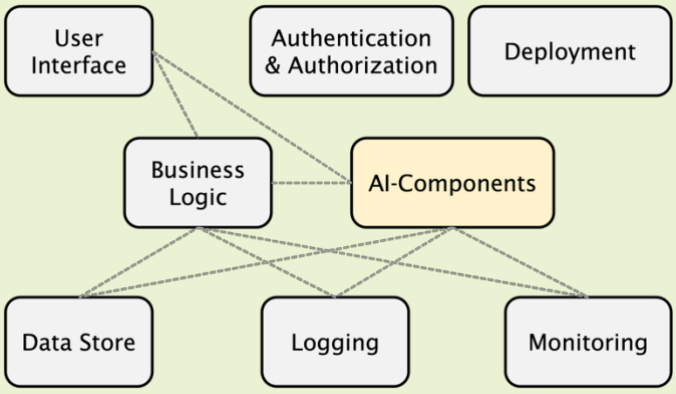
AGREE

COMPLETELY AGREE

Please answer the questions taking the following architectural rule into account:

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

The figure below illustrates the architecture of an AI-based system. It consists of one or more **AI components** grouped into a single module and some additional supportive **non-AI components** (e.g., user interfaces and business logic).



- **AI-Component(s)**: Algorithms (including 3rd party libraries) used for training, testing and validating ML models.
- **User interface**: Responsible for gathering input from the users and displaying results.
- **Business logic**: Responsible for implementing rules that define how data should be processed, managed, and controlled.
- **Authentication and authorization**: Ensores that only those who have the right permissions can access the system and its components.
- **Deployment**: Responsible for integrating trained ML models to an existing production environment.
- **Datastore**: Contains the data employed for training, testing, and validating ML models.
- **Logging**: Captures information about events and actions within the ML system (e.g., prediction outputs and error messages) that can be used for in-depth analysis and debugging.

- **Monitoring**: Provides a real-time overview of the performance and health of the ML models (e.g., model accuracy and resource consumption) to ensure that they are running as expected.

Question attribute	Value
save_as_default	N

Based on the architecture described above and the following architectural rule:

► RULE 1: "Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed"

For which component(s) of an AI-based system architecture do you consider this rule relevant?

(Tip: Select all that apply)

Question attribute	Value
other_position	end
save_as_default	N

AI-Component(s)

User interface

Business logic

Authentication and authorization

Deployment

Datastore

Logging

Monitoring

Please answer the questions taking the following architectural rule into account:

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

Based on your experience, please indicate **how often** you have applied this **architectural rule** while working with AI systems.

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

NEVER

RARELY

SOMETIMES

OFTEN

ALWAYS

Based on your own experience, how could you check if an AI system **conforms to this rule** (i.e., does not violate it)?

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify
other_position	end
other	Y
save_as_default	N

Manually, following a structured or systematic approach (e.g., checklists).

Manually, not following any structured or systematic approach (intuitively).

(Semi-) automatically, tool-supported (e.g., static analysis).

Other

Which **sources/types of information** could you use to check this rule?

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify.
other_position	end
other	Y
save_as_default	N

Abstract representations of the system architecture (e.g., data flow diagrams).

Unstructured architectural information available across artefacts (e.g., code comments, code review comments).

Third-party documentation (e.g., API documentation).

Structured architectural information available as official documentation

Other

Can you explain how the **conformance check** for this rule would be **conducted**? (e.g., the overall strategy, use of sources, etc.)

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

Question attribute	Value

save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

Please indicate, based on your experience working with AI systems, how much do you **agree** or **disagree** with the following statements.

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

Question attribute	Value
save_as_default	N

I believe this rule helps create AI architectures with clearly defined and easy-to-maintain modules

I believe this rule helps reduce inconsistencies (e.g., unnecessary steps and errors) across the deployment pipeline of AI systems

I believe this rule contributes to the effective testing of AI applications and facilitates their transferability (e.g., handing them over to colleagues)

I believe this rule helps avoid unintended/unwanted accesses to the output of an AI module (e.g., from another AI module)

I believe this rule helps avoid dependency chains across AI models and the need for hotfixes to improve the accuracy of the final model

I believe this rule helps reduce the amount of additional/supporting code needed to get data into and out of AI modules (e.g., packages)

I believe this rule helps avoid having multiple AI components packed into a single big file

I believe this rule helps reduce the number of fixes/adjustments across multiple AI components when 3rd-party libraries get updated

I believe this rule helps reduce the number of unnecessary code fragments, methods or classes added to debug AI systems

COMPLETELY DISAGREE

DISAGREE

SOMEHOW DISAGREE

SOMEHOW AGREE

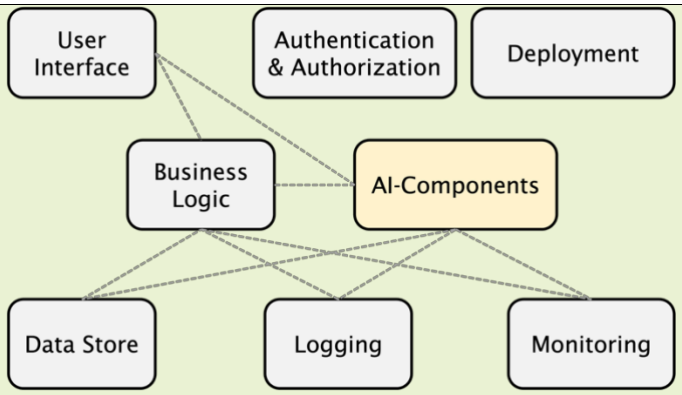
AGREE

COMPLETELY AGREE

Please answer the questions taking the following architectural rule into account:

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

The figure below illustrates the architecture of an AI-based system. It consists of one or more **AI components** grouped into a single module and some additional supportive **non-AI components** (e.g., user interfaces and business logic).



- **AI-Component(s)**: Algorithms (including 3rd party libraries) used for training, testing and validating ML models.
- **User interface**: Responsible for gathering input from the users and displaying results.
- **Business logic**: Responsible for implementing rules that define how data should be processed, managed, and controlled.
- **Authentication and authorization**: Ensores that only those who have the right permissions can access the system and its components.
- **Deployment**: Responsible for integrating trained ML models to an existing production environment.
- **Datastore**: Contains the data employed for training, testing, and validating ML models.
- **Logging**: Captures information about events and actions within the ML system (e.g., prediction outputs and error messages) that can be used for in-depth analysis and debugging.
- **Monitoring**: Provides a real-time overview of the performance and health of the ML models (e.g., model accuracy and resource consumption) to ensure that they are running as expected.

Question attribute	Value
save_as_default	N

Based on the architecture described above and the following architectural rule:

► RULE 2: "Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance"

For which component(s) of an AI-based system architecture do you consider this rule relevant?

(Tip: Select all that apply)

Question attribute	Value
other_position	end
save_as_default	N

AI-Component(s)

User interface

Business logic

Authentication and authorization

Deployment

Datastore

Logging

Monitoring

Please answer the questions taking the following architectural rule into account:

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

Based on your experience, please indicate **how often** you have applied this **architectural rule** while working with AI systems.

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

NEVER

RARELY

SOMETIMES

OFTEN

ALWAYS

Based on your own experience, how could you check if an AI system **conforms to this rule** (i.e., does not violate it)?

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify
other_position	end
other	Y
save_as_default	N

Manually, following a structured or systematic approach (e.g., checklists).

Manually, not following any structured or systematic approach (intuitively).

(Semi-) automatically, tool-supported (e.g., static analysis).

Other

Which **sources/types of information** could you use to check this rule?

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify.

other_position	end
other	Y
save_as_default	N

Abstract representations of the system architecture (e.g., data flow diagrams).

Unstructured architectural information available across artefacts (e.g., code comments, code review comments).

Third-party documentation (e.g., API documentation).

Structured architectural information available as official documentation

Other

Can you explain how the **conformance check** for this rule would be **conducted**? (e.g., the overall strategy, use of sources, etc.)

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

Question attribute	Value
save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

Please indicate, based on your experience working with AI systems, how much do you **agree** or **disagree** with the following statements.

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

Question attribute	Value
save_as_default	N

I believe this rule helps create AI architectures with clearly defined and easy-to-maintain modules

I believe this rule helps reduce inconsistencies (e.g., unnecessary steps and errors) across the deployment pipeline of AI systems

I believe this rule contributes to the effective testing of AI applications and facilitates their transferability (e.g., handing them over to colleagues)

I believe this rule helps avoid unintended/unwanted accesses to the output of an AI module (e.g., from another AI module)

I believe this rule helps avoid dependency chains across AI models and the need for hotfixes to improve the accuracy of the final model

I believe this rule helps reduce the amount of additional/supporting code needed to get data into and out of AI modules (e.g., packages)

I believe this rule helps avoid having multiple AI components packed into a single big file

I believe this rule helps reduce the number of fixes/adjustments across multiple AI components when 3rd-party libraries get updated

I believe this rule helps reduce the number of unnecessary code fragments, methods or classes added to debug AI systems

COMPLETELY DISAGREE

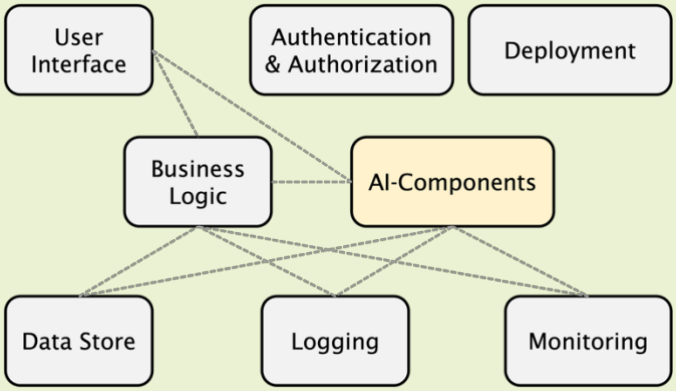
DISAGREE

SOMEHOW DISAGREE
SOMEHOW AGREE
AGREE
COMPLETELY AGREE

Please answer the questions taking the following architectural rule into account:

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

The figure below illustrates the architecture of an AI-based system. It consists of one or more **AI components** grouped into a single module and some additional supportive **non-AI components** (e.g., user interfaces and business logic).



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- **Deployment**: Responsible for integrating trained ML models to an existing production environment.
- **Datastore**: Contains the data employed for training, testing, and validating ML models.
- **Logging**: Captures information about events and actions within the ML system (e.g., prediction outputs and error messages) that can be used for in-depth analysis and debugging.
- **Monitoring**: Provides a real-time overview of the performance and health of the ML models (e.g., model accuracy and resource consumption) to ensure that they are running as expected.

Question attribute	Value
save_as_default	N

Based on the architecture described above and the following architectural rule:

► RULE 3: "Continuously check the outcomes of AI components against ethical requirements (e.g., fairness)"

For which component(s) of an AI-based system architecture do you consider this rule relevant?

(Tip: Select all that apply)

Question attribute	Value

other_position	end
save_as_default	N

AI-Component(s)

User interface

Business logic

Authentication and authorization

Deployment

Datastore

Logging

Monitoring

Please answer the questions taking the following architectural rule into account:

► RULE 4: "Continuously check for discrepancies across redundant AI components"

Based on your experience, please indicate **how often** you have applied this **architectural rule** while working with AI systems.

► RULE 4: "Continuously check for discrepancies across redundant AI components"

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

NEVER

RARELY

SOMETIMES

OFTEN

ALWAYS

Based on your own experience, how could you check if an AI system **conforms to this rule** (i.e., does not violate it)?

► RULE 4: "Continuously check for discrepancies across redundant AI components"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify
other_position	end
other	Y
save_as_default	N

Manually, following a structured or systematic approach (e.g., checklists).

Manually, not following any structured or systematic approach (intuitively).

(Semi-) automatically, tool-supported (e.g., static analysis).

Other

Which **sources/types of information** could you use to check this rule?

► RULE 4: "Continuously check for discrepancies across redundant AI components"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify.
other_position	end
other	Y
save_as_default	N

Abstract representations of the system architecture (e.g., data flow diagrams).

Unstructured architectural information available across artefacts (e.g., code comments, code review comments).

Third-party documentation (e.g., API documentation).

Structured architectural information available as official documentation

Other

Can you explain how the **conformance check** for this rule would be **conducted**? (e.g., the overall strategy, use of sources, etc.)

► RULE 4: "Continuously check for discrepancies across redundant AI components"

Question attribute	Value
save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 4: "Continuously check for discrepancies across redundant AI components"

Please indicate, based on your experience working with AI systems, how much do you **agree** or **disagree** with the following statements.

► RULE 4: "Continuously check for discrepancies across redundant AI components"

Question attribute	Value
save_as_default	N

I believe this rule helps create AI architectures with clearly defined and easy-to-maintain modules

I believe this rule helps reduce inconsistencies (e.g., unnecessary steps and errors) across the deployment pipeline of AI systems

I believe this rule contributes to the effective testing of AI applications and facilitates their transferability (e.g., handing them over to colleagues)

I believe this rule helps avoid unintended/unwanted accesses to the output of an AI module (e.g., from another AI module)

I believe this rule helps avoid dependency chains across AI models and the need for hotfixes to improve the accuracy of the final model

I believe this rule helps reduce the amount of additional/supporting code needed to get data into and out of AI modules (e.g., packages)

I believe this rule helps avoid having multiple AI components packed into a single big file

I believe this rule helps reduce the number of fixes/adjustments across multiple AI components when 3rd-party libraries get updated

I believe this rule helps reduce the number of unnecessary code fragments, methods or classes added to debug AI systems

COMPLETELY DISAGREE

DISAGREE

SOMEHOW DISAGREE

SOMEHOW AGREE

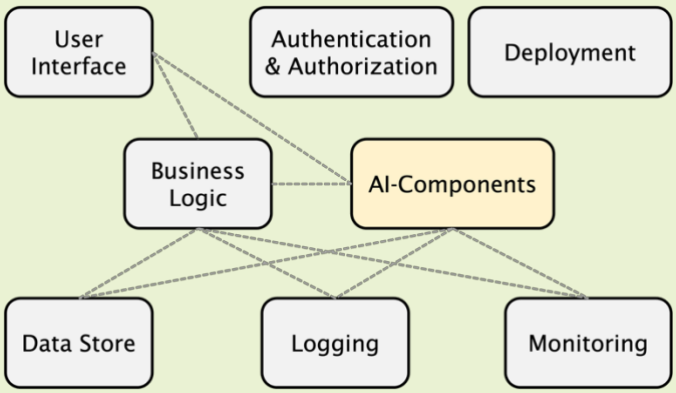
AGREE

COMPLETELY AGREE

Please answer the questions taking the following architectural rule into account:

► RULE 4: "Continuously check for discrepancies across redundant AI components"

The figure below illustrates the architecture of an AI-based system. It consists of one or more **AI components** grouped into a single module and some additional supportive **non-AI components** (e.g., user interfaces and business logic).



- **AI-Component(s)**: Algorithms (including 3rd party libraries) used for training, testing and validating ML models.
- **User interface**: Responsible for gathering input from the users and displaying results.
- **Business logic**: Responsible for implementing rules that define how data should be processed, managed, and controlled.
- **Authentication and authorization**: Ensores that only those who have the right permissions can access the system and its components.
- **Deployment**: Responsible for integrating trained ML models to an existing production environment.
- **Datastore**: Contains the data employed for training, testing, and validating ML models.
- **Logging**: Captures information about events and actions within the ML system (e.g., prediction outputs and error messages) that can be used for in-depth analysis and debugging.
- **Monitoring**: Provides a real-time overview of the performance and health of the ML models (e.g., model accuracy and resource consumption) to ensure that they are running as expected.

Question attribute	Value
save_as_default	N

Based on the architecture described above and the following architectural rule:

► RULE 4: "Continuously check for discrepancies across redundant AI components"

For which component(s) of an AI-based system architecture do you consider this rule relevant?

(Tip: Select all that apply)

Question attribute	Value
other_position	end
save_as_default	N

- AI-Component(s)
- User interface
- Business logic
- Authentication and authorization
- Deployment
- Datastore
- Logging
- Monitoring

Please answer the questions taking the following architectural rule into account:

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

Based on your experience, please indicate **how often** you have applied this **architectural rule** while working with AI systems.

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

(Tip: Choose one of the following answers)

Question attribute	Value
answer_order	normal
other_position	default
save_as_default	N

- NEVER
- RARELY
- SOMETIMES
- OFTEN
- ALWAYS

Based on your own experience, how could you check if an AI system **conforms to this rule** (i.e., does not violate it)?

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify
other_position	end
other	Y
save_as_default	N

Manually, following a structured or systematic approach (e.g., checklists).

Manually, not following any structured or systematic approach (intuitively).

(Semi-) automatically, tool-supported (e.g., static analysis).

Other

Which **sources/types of information** could you use to check this rule?

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

(Tip: Select all that apply)

Question attribute	Value
other_replace_text	Other: Please specify.
other_position	end
other	Y
save_as_default	N

Abstract representations of the system architecture (e.g., data flow diagrams).

Unstructured architectural information available across artefacts (e.g., code comments, code review comments).

Third-party documentation (e.g., API documentation).

Structured architectural information available as official documentation

Other

Can you explain how the **conformance check** for this rule would be **conducted**? (e.g., the overall strategy, use of sources, etc.)

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

Question attribute	Value
save_as_default	N

Please answer the questions taking the following architectural rule into account:

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

Please indicate, based on your experience working with AI systems, how much do you **agree** or **disagree** with the following statements.

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

Question attribute	Value
save_as_default	N

I believe this rule helps create AI architectures with clearly defined and easy-to-maintain modules

I believe this rule helps reduce inconsistencies (e.g., unnecessary steps and errors) across the deployment pipeline of AI systems

I believe this rule contributes to the effective testing of AI applications and facilitates their transferability (e.g., handing them over to colleagues)

I believe this rule helps avoid unintended/unwanted accesses to the output of an AI module (e.g., from another AI module)

I believe this rule helps avoid dependency chains across AI models and the need for hotfixes to improve the accuracy of the final model

I believe this rule helps reduce the amount of additional/supporting code needed to get data into and out of AI modules (e.g., packages)

I believe this rule helps avoid having multiple AI components packed into a single big file

I believe this rule helps reduce the number of fixes/adjustments across multiple AI components when 3rd-party libraries get updated

I believe this rule helps reduce the number of unnecessary code fragments, methods or classes added to debug AI systems

COMPLETELY DISAGREE

DISAGREE

SOMEHOW DISAGREE

SOMEHOW AGREE

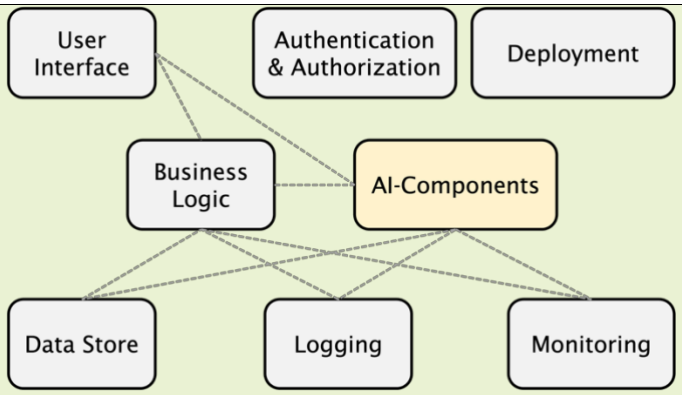
AGREE

COMPLETELY AGREE

Please answer the questions taking the following architectural rule into account:

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

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Question attribute	Value
save_as_default	N

Based on the architecture described above and the following architectural rule:

► RULE 5: "Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties"

For which component(s) of an AI-based system architecture do you consider this rule relevant?

(Tip: Select all that apply)

Question attribute	Value
other_position	end
save_as_default	N

AI-Component(s)

User interface

Business logic

Authentication and authorization

Deployment

Datastore

Logging

Monitoring

Throughout this survey, you have assessed the use of the following architectural rules for securing AI-based systems:

- **R1**: Implement AI mode switchers that allow users (humans) to activate/deactivate AI components or override their decisions when needed
- **R2**: Implement decision-making modules through multiple/different AI components to increase the system's reliability and fault-tolerance.
- **R3**: Continuously check the outcomes of AI components against ethical requirements (e.g., fairness).
- **R4**: Continuously check for discrepancies across redundant AI components.
- **R5**: Create a Bill of Materials (BoM) registry to keep a machine-readable record of all used components and their properties.

Is there any rule (other than these ones) that you would consider applying to secure AI-based system architectures? (answer "none" if not)

Question attribute	Value
save_as_default	N

Thank you for your interest in our survey. Unfortunately, you do not meet some of the requirements to participate in this study:

- Be at least 18 years old.
- Read the consent form or had it read to you.
- Be comfortable using the English language.
- Agree to participate in this research.

If you have any questions or inquiries, please do not hesitate to contact us a [redacted]

Thank you for your response!

IMPORTANT: Please click **"SUBMIT"** to record your answer!

Question attribute	Value
save_as_default	N